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VETERINARY MICROBIOLOGY · RAPID REVISION

Unit 5 — General & Systematic Veterinary Virology

All 33 syllabus topics on six working sheets. The systematic half is organised the way the examiner thinks — by *genome type and family*, not alphabetically — so that once you know which family a virus belongs to, its structure, its diagnosis and its vaccine follow automatically.

PAPER
Paper II · Units 4, 5
Theory 100 · Practical 60

SYLLABUS TOPICS
33 of 33 covered

REVISION TIME
30 min full pass
14 min sheets 03–06

COURSE
B.V.Sc & A.H. II Yr
VCI MSVE · 3+2=5

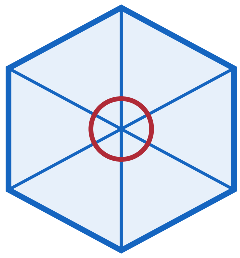
General virology · Replication & cultivation · RNA viruses · DNA viruses · Prions & emerging · Exam trap & mnemonic

What a Virus Is — Structure, Classification, Cultivation

Virion architecture & symmetry

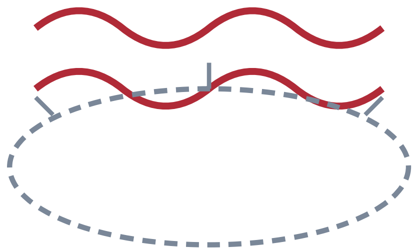
MECHANISM

ICOSAHEDRAL (cubical)



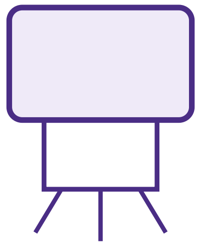
20 triangular faces, 12 vertices
Capsomeres = hexons + 12 pentons
T number: capsomeres = 10T + 2
Adeno-, herpes-, parvo-, picorna-, reo-, circo-, papillomaviruses

HELICAL



Nucleic acid coiled inside a rod of identical protomers.
In animal viruses always **ENVELOPED** — paramyxo-, ortho-, rhabdo-, corona-, bornaviruses

COMPLEX



Neither purely cubical nor helical. **Poxvirus** (brick-shape dumbbell core, lateral bodies — **largest animal virus**) and the tailed bacteriophages

- **Virion** = the complete infectious particle. **Capsid** = protein coat made of **capsomeres**, themselves built of **protomers**. **Nucleocapsid** = genome + capsid. **Envelope** = host-derived lipid bilayer with virus-coded **peplomers/spikes**. **Tegument** = the protein layer between capsid and envelope in herpesviruses.
- **Enveloped vs naked — the practical consequences.** Enveloped: **ether- and chloroform-sensitive**, destroyed by detergents, soap, drying and heat, so they spread by close contact and are killed by ordinary disinfectants. Naked: **resistant** to lipid solvents, acid and drying, survive in the environment, spread by the faecal-oral route and need strong disinfectants (FMD, parvo, circo, IBD).
- **Size:** 20 nm (circovirus, the smallest) to 300–450 nm (poxvirus). Pass **0.22 µm filters**; visible only by **electron microscopy**.

General properties & how a virus differs

TABLE

Feature	Virus	Bacterium
Nucleic acid	**DNA or RNA, never both**	Both
Ribosomes	**Absent**	70S present
Metabolism	None — **obligate intracellular parasite at the genetic level**	Own metabolism
Multiplication	**Eclipse phase, then assembly** — not binary fission	Binary fission
Growth on inert media	**No**	Yes (except *Rickettsia*, *Chlamydia*)
Filterable	Yes	No
Antibiotics	**Insensitive**	Sensitive
Interferon	Sensitive	Insensitive
Crystallisable	**Yes** (TMV, poliovirus)	No

History 1892 Iwanowski — filterable agent of tobacco mosaic · **1898 Beijerinck** coined "contagium vivum fluidum" and the word *virus* · **1898 Loeffler & Frosch — FMD, the first ANIMAL virus** · **1915–17 Twort & d'Herelle** — bacteriophage · **1935 Stanley** crystallised TMV · 1931 Woodruff & Goodpasture — egg cultivation · 1949 Enders, Weller & Robbins — poliovirus in cell culture (Nobel) · **1982 Prusiner — prions**.

Sub-viral agents you must be able to define: Viroid — naked circular ssRNA, no protein, plant pathogens · **Virusoid/satellite** — needs a helper virus · **Prion** — infectious protein only, no nucleic acid · **Defective interfering (DI) particle** — incomplete genome, interferes with the parent virus · **Pseudovirion** — host DNA inside a viral capsid.

Classification — the Baltimore scheme & the family table

MECHANISM

EVERY VIRUS MUST MAKE mRNA — Baltimore classifies by HOW

- **I · dsDNA** — herpes, adeno, pox, asfar, papilloma,
- **II · ssDNA** — parvo, circo (Anello)
- **VII · dsDNA-RT** — hepadna (hepatitis B)
- **III · dsRNA** — reo, birna (segmented)
- **IV · (+)ssRNA** — picorna, calici, toga, flavi, coror
- **V · (−)ssRNA** — paramyxo, rhabdo, orthomyxo, l
- **VI · (+)ssRNA-RT** — retro (lenti, deltaretro, alpha

Key consequence: groups V and VI must carry their own polymerase in the virion

- **ICTV hierarchy:** Order (**-virales**) → Family (**-viridae**) → Subfamily (**-virinae**) → Genus (**-virus**) → Species. Family, genus and species names are **italicised**; the vernacular name is not.
- **Basis of classification:** nature of the nucleic acid (DNA/RNA, ss/ds, ±sense, segmented or not), **symmetry**, **presence of an envelope**, size, site of replication, and now genome sequence.
- **Older systems worth naming:** Lwoff-Horne-Tournier (LHT) system based on nucleic acid, symmetry and envelope; Baltimore (1971) as above.
- **Two rules worth memorising because they answer many objective questions:** (1) **All DNA viruses replicate in the nucleus except poxvirus and African swine fever virus**, which replicate in the cytoplasm. (2) **All RNA viruses replicate in the cytoplasm except orthomyxovirus (influenza) and the retroviruses**, which need the nucleus.

- Another pair: All DNA viruses are double-stranded except parvovirus and circovirus; all RNA viruses are single-stranded except reovirus and birnavirus.

Cultivation of viruses

TABLE

System	Detail worth writing
Laboratory animals	Suckling mice by the **intracerebral** route for rabies and arboviruses; guinea pigs, rabbits, hamsters. Used for pathogenicity studies, antisera and where no cell line supports the virus. Ethically restricted — the last resort
Embryonated chicken egg `Goodpasture & Woodruff, 1931`	**The four routes and what each is for:** **Chorioallantoic membrane (CAM), 11–12 d** → discrete **pocks** — pox viruses, IBR, laryngotracheitis; pock morphology distinguishes strains **Allantoic cavity, 9–11 d** → highest virus yield — **Newcastle disease and influenza vaccine production** **Amniotic cavity, 12–15 d** → primary isolation of influenza **Yolk sac, 5–7 d** → *Chlamydia*, *Rickettsia*, some herpesviruses Advantages: sterile, cheap, no immune response, wide susceptibility. Read as pocks, death, haemorrhage, stunting, curling or dwarfing of the embryo
Cell culture	**Primary** (chick embryo fibroblast, lamb/calf kidney — a few passages only, best for isolation) → **diploid/semi-continuous** (finite ~50 passages, retain karyotype) → **continuous cell lines** (immortal, heteroploid — **Vero, BHK-21, MDBK, MDCK, HeLa, PK-15, CRFK**). Cells grow as a monolayer requiring **anchorage and contact inhibition**, or in suspension
Media & conditions	**MEM/DMEM/Medium 199** + **10% foetal bovine serum** + antibiotics + **phenol red** as pH indicator + sodium bicarbonate; 37 °C in 5% CO_2. **Trypsin-EDTA** to detach cells for subculture; **trypan blue** for viable count in a haemocytometer; **cryopreservation in 10% DMSO or glycerol, slow freeze, fast thaw**, stored in liquid nitrogen

Quantitation **Infectivity assays** — **$TCID_{50}$** calculated by the **Reed-Muench** or Kärber method, **plaque assay (PFU/ml)** under an agar overlay with neutral red, **EID_{50}** in eggs, LD_{50} in animals. **Physical assays** — haemagglutination titre (HA units), electron-microscope particle count, ELISA and qPCR copy number. Physical counts are always higher than infectivity counts — the **particle-to-infectivity ratio**.

Sheet 01 · 60-second self-test

RECALL

First animal virus discovered	FMD virus — Loeffler & Frosch, 1898
Coined the word virus	Beijerinck
Largest animal virus	Poxvirus (brick-shaped)
Smallest animal virus	Circovirus
Capsomere number formula	10T + 2
Faces of an icosahedron	20 faces, 12 vertices
Helical animal viruses are always	Enveloped
Ether-sensitive viruses	Enveloped ones
DNA viruses replicating in cytoplasm	Pox and African swine fever
RNA viruses needing the nucleus	Orthomyxo and retroviruses
Single-stranded DNA viruses	Parvo and circovirus
Double-stranded RNA viruses	Reo and birnavirus
Route for pock production	Chorioallantoic membrane
Route for ND vaccine production	Allantoic cavity, 9–11 day egg
Yolk sac is used for	*Chlamydia* and *Rickettsia*
Cell line from African green monkey	Vero
Enzyme used to subculture cells	Trypsin-EDTA
Cryoprotectant for cell lines	10% DMSO / glycerol
Method to compute $TCID_{50}$	Reed-Muench
Infectious protein without nucleic acid	Prion

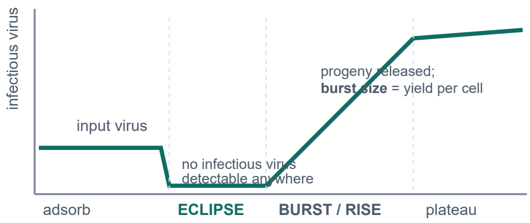
02 Replication, Interactions, CPE & Pathogenesis

t04–t07

The replication cycle and the one-step growth curve

MECHANISM

- **1 ADSORPTION** — Ligand meets a specific receptor decides tropism
- **2 PENETRATION** — Endocytosis (nak or membrane fus (enveloped)
- **3 UNCOATING** — Capsid removed; genome released **ECLIPSE begins**
- **4 BIOSYNTHESI** — Early (non-structu → genome copie late (structural) p
- **5 ASSEMBLY** — Nucleus (most DN or cytoplasm; ma form inclusion bo
- **6 RELEASE** — **Lysis** — naked **Budding** — env cell may survive



LATENT PERIOD = uncoating to first extracellular virus. **ECLIPSE** = unco

Genetic & non-genetic interactions

TABLE

Interaction	What happens & the standard example
GENETIC — the change is heritable	
Mutation	Point, deletion, insertion. RNA viruses mutate ~1000× faster because **RNA polymerase has no proofreading** → quasispecies. Basis of attenuation and of **antigenic drift**
Recombination	Exchange of nucleotide sequence between two related genomes — occurs in DNA viruses and in **picornaviruses and coronaviruses**
Reassortment	Exchange of whole **segments** when two strains infect one cell — only in **segmented** genomes: influenza, rotavirus, bluetongue. **This is the mechanism of ANTIGENIC SHIFT and of pandemic influenza**
Complementation	One virus supplies a functional protein that the other lacks; **genomes are unchanged** — a defective virus is rescued by a helper
Cross-reactivation (marker rescue)	An active virus rescues a gene from an inactivated one
Multiplicity reactivation	Several inactivated particles, each damaged in a different gene, together produce infectious progeny
NON-GENETIC — phenotype only, not inherited	
Phenotypic mixing	Genome of one virus inside the capsid or envelope of another; progeny revert to the parental type
Phenotypic masking (transcapsidation)	Genome completely enclosed in the other's capsid
Genome masking	Same idea, complete coat exchange
Interference	Infection with one virus prevents another — mediated by **interferon**, receptor blockade or **defective interfering particles**. Practical importance: interference between live vaccine viruses given together

The distinction that carries the mark: recombination exchanges parts of genes and can occur in any genome; reassortment swaps entire genome segments and can only occur in a segmented virus. Antigenic drift = accumulated point mutations, gradual, causes seasonal epidemics; antigenic shift = reassortment, abrupt, causes pandemics.

Virus-cell interactions & cytopathic effects

TABLE

Outcome	Description & example
Lytic (cytocidal)	Cell dies and releases virus — enteroviruses, FMD
Persistent / chronic	Continuous low-level production, cell survives — BVDV, EIA
Latent	Genome present, **no infectious virus made**, reactivates on stress or corticosteroid — **herpesviruses (IBR in the trigeminal ganglion, pseudorabies, Marek's)**
Transforming	Cell becomes immortal, loses contact inhibition, piles up in foci — oncogenic viruses
Abortive / non-productive	Non-permissive cell; the cycle starts but no progeny

CPE type	Virus
Rounding, shrinkage, detachment	Picornaviruses, FMD — the commonest CPE
Syncytium / polykaryon (giant multinucleate cell)	**Paramyxoviruses — PPR, CDV, NDV, RSV, BVDV cytopathic biotype**
Grape-like clusters	Adenovirus
Cell fusion + intranuclear inclusion	Herpesvirus
Vacuolation, foamy	SV40, some retroviruses
Haemadsorption	Influenza, parainfluenza — RBCs stick to infected cells before visible CPE
No CPE	Rabies, non-cytopathic BVDV — must be detected by FAT or immunoperoxidase

Inclusion bodies — memorise the six **Negri body** — rabies, **intracytoplasmic, eosinophilic**, in Purkinje cells and hippocampal neurons · **Bollinger body** — fowl pox, intracytoplasmic · **Guarnieri body** — vaccinia/variola, intracytoplasmic · **Cowdry type A** — herpes, **intranuclear**; also in ICH (adenovirus) · **Intranuclear basophilic** — canine adenovirus 1, IBR · **Both intranuclear and intracytoplasmic** — canine distemper (the classic "both" answer).

Viral pathogenesis

RAPID-FIRE

Sequence to write

Entry & **primary replication** at the portal → **primary viraemia** → replication in liver, spleen, endothelium → **secondary viraemia** → localisation in the **target organ** → shedding → outcome

Routes of entry	Respiratory, alimentary, skin abrasion or bite, arthropod vector, venereal, conjunctival, iatrogenic, **vertical/transplacental**
Spread in the body	**Local** (papilloma, pox) · **Haematogenous**, free in plasma or cell-associated · **Neural — rabies travelling by retrograde axonal transport at 3 mm/h** · lymphatic
Tropism is decided by	The **receptor** (CD4 for HIV, SLAM/nectin-4 for morbilliviruses, sialic acid for influenza, integrin for FMD), plus intracellular factors and route of entry
Mechanisms of cell injury	Shut-off of host protein synthesis, lysosomal release, inclusion formation, apoptosis, membrane damage, and **immunopathology**
Immunopathology to name	Type III complexes → **blue eye of ICH**, glomerulonephritis of **Aleutian disease of mink** and EIA · **Antibody-dependent enhancement** (FIP, dengue) · Cytotoxic T-cell mediated damage · **Immunosuppression — BVDV, CDV, IBDV/Gumboro, FIV, ASFV, PCV-2**
Shedding	Respiratory aerosol, faeces, urine (**leptospira-like shedding of ICH virus for 6 months**), saliva, milk, semen, skin scales

Latency & viral oncogenesis

RAPID-FIRE

Latency — the model answer	Herpesviral genome persists as a **circular episome** in a non-dividing cell (sensory ganglion) with only latency-associated transcripts expressed. **No virion, no antigen → invisible to the immune system.** Reactivated by stress, transport, parturition or **corticosteroid (dexamethasone — the experimental trigger)** → recrudescence and shedding. **IBR in the trigeminal ganglion is the standard veterinary example**; the animal is a lifelong carrier, which is why **seropositive animals are culled in eradication programmes**
Oncogenic mechanisms	**1 Insertional mutagenesis** — a retroviral LTR promoter inserts near a cellular proto-oncogene and switches it on (**avian leucosis virus and *c-myc*; bovine leukaemia virus**) · **2 Transduced viral oncogene (*v-onc*)** — acutely transforming retroviruses such as Rous sarcoma virus and *v-src* · **3 Viral proteins inactivating tumour suppressors** — papillomavirus E6 and E7 binding **p53 and Rb** · **4** Chronic inflammation and regeneration
Oncogenic viruses in veterinary practice	**Marek's disease virus** (herpes — T-cell lymphoma in chickens, the **first cancer prevented by a vaccine**) · **Avian leucosis/sarcoma viruses** · **Bovine leukaemia virus** (enzootic bovine leucosis) · **Feline leukaemia virus** · **Bovine and canine papillomaviruses** (BPV-1/2 with bracken fern → bovine bladder cancer and equine sarcoid) · Ovine pulmonary adenocarcinoma (jaagsiekte retrovirus) · Rous sarcoma virus, the historic first (Peyton Rous, 1911)
Marks of a transformed cell	Loss of **contact inhibition**, anchorage independence, immortality, altered surface antigens, aneuploidy, growth in soft agar, tumour formation in a nude mouse

Interferon in one box: type I (α from leucocytes, β from fibroblasts) is induced by **dsRNA**, is **species-specific but not virus-specific**, acts on neighbouring cells to induce an **antiviral state** through **2′-5′ oligoadenylate synthetase/RNase L and PKR** which degrade mRNA and block translation. **Type II (IFN-γ)** is immunological rather than directly antiviral. Interferon is the **earliest antiviral defence**, appearing before antibody.

Birnaviridae & Reoviridae — the dsRNA block

SYSTEMATIC

- **Infectious bursal disease virus (Gumboro disease) — *Birnaviridae*. Bi-segmented dsRNA, naked icosahedral**, extremely **resistant** in the environment (survives 122 days in a house). Targets **immature IgM-bearing B lymphocytes in the bursa of Fabricius. Age 3–6 weeks** is the classic window. Signs: sudden onset, prostration, vent picking, whitish watery diarrhoea, **spiking mortality that recovers in 5–7 days**. Lesions: **bursa first swollen, oedematous and gelatinous with a yellow transudate, then atrophic by day 8; haemorrhages in thigh and pectoral muscles**; nephrosis, "chalky" urates. **The lasting damage is IMMUNOSUPPRESSION** — vaccine failures against ND and Marek's follow. Diagnosis: AGPT of bursa, ELISA, RT-PCR, histopathology. Control: **intermediate and intermediate-plus live vaccines timed by the Deventer formula against maternal antibody, plus breeder vaccination**; strict biosecurity and formaldehyde fumigation.
- ***Reoviridae* — segmented (10–12), dsRNA, naked, double capsid.**
- **Bluetongue virus** — *Orbivirus*, **10 segments**, >27 serotypes, transmitted by ***Culicoides* midges** (biological vector) → **seasonal, vector-linked, non-contagious**. Sheep are worst affected. Signs: fever, facial and lip oedema, hyperaemic muzzle, **cyanotic swollen protruding tongue**, coronitis with a red band above the hoof, wry neck, foetal hydranencephaly. Lesion: **haemorrhage at the base of the pulmonary artery — pathognomonic**. Diagnosis: **AGID and c-ELISA** for group antigen, **serotyping by SNT**, RT-PCR, isolation in embryonated eggs by **intravenous** inoculation or in BHK-21/Vero. **Notifiable, transboundary.**
- **African horse sickness virus** — *Orbivirus*, 9 serotypes, also *Culicoides*-borne. Four forms: **pulmonary (dunkop — peracute, frothy nasal discharge, ~95% mortality), cardiac (dikkop — supraorbital fossa oedema)**, mixed, and horse-sickness fever. Zebras are the reservoir. Notifiable.
- **Rotavirus — wheel-like appearance on EM**, 11 segments, groups A–G (**group A causes most neonatal diarrhoea**). Infects the **mature enterocytes on the villus tips of the small intestine** → villous atrophy, malabsorptive and osmotic diarrhoea in calves, piglets, lambs and foals in the **first 1–2 weeks**, usually with cryptosporidia, coronavirus and ETEC. Diagnosis: **EM, ELISA, latex agglutination, PAGE of segments (electropherotyping), RT-PCR**. Control: **vaccinate the dam in late gestation so that colostral and milk antibody protects the gut lumen** — a favourite question.

Paramyxoviridae — Newcastle disease

SYSTEMATIC

- **Structure:** enveloped, **helical, non-segmented (−)ssRNA**, pleomorphic. Two glycoprotein spikes: **HN (haemagglutinin-neuraminidase)** and **F (fusion) protein. Cleavage of the F_0 precursor by host proteases determines virulence** — a polybasic cleavage site allows cleavage by ubiquitous proteases and therefore systemic spread. **APMV-1 — a single serotype**, which is why any vaccine strain protects against any field strain.
- **Pathotypes: Velogenic** (viscerotropic = Doyle's form, haemorrhagic gut; neurotropic = Beach's form) · **Mesogenic** (Beaudette's, respiratory + nervous, moderate) · **Lentogenic** (Hitchner B1, **LaSota**, F strain — mild, used as vaccines) · asymptomatic enteric.
- **Signs:** respiratory distress and gasping, greenish diarrhoea, **nervous signs — torticollis, paralysis, opisthotonus, circling**, marked **drop in egg production with soft-shelled and misshapen eggs**, swelling around the eyes and head, high mortality.
- **Lesions: haemorrhagic and necrotic ulcers over the lymphoid tissue of the gut — Peyer's patches, caecal tonsils, petechiae on the proventricular gland tips**, haemorrhagic tracheitis, ovarian follicle degeneration.
- **Diagnosis:** isolation by **allantoic-cavity inoculation of 9–11 day eggs** → harvest allantoic fluid → **HA test to detect, HI test with specific antiserum to identify**; pathotyping by **MDT (mean death time in eggs — velogenic <60 h, mesogenic 60–90 h, lentogenic >90 h), ICPI (intracerebral pathogenicity index in day-old chicks — velogenic approaches 2.0)**, and F-protein cleavage-site sequencing; RT-PCR and ELISA.
- **Control:** in India, **F1 (Hitchner B1) strain at 5–7 days by eye drop, LaSota booster at 4–5 weeks, and R2B (Mukteswar mesogenic strain) at 8–10 weeks by subcutaneous route**; killed oil-adjuvant vaccine for layers and breeders. Monitor by **HI titre**. Notifiable and transboundary.

> Ranikhet disease is the Indian name for Newcastle disease — say both. Zoonotic note: causes a self-limiting conjunctivitis in poultry workers and vaccinators.

Morbilliviruses — PPR & canine distemper

TABLE

	Peste des petits ruminants (PPR)	Canine distemper (CDV)
Host	**Goats > sheep**; goats more severely affected	Dog, fox, ferret, mink, lion, **seal**; young unvaccinated pups 3–6 months
Also called	**Goat plague**, pseudo-rinderpest, kata	Carré's disease, hard pad disease
Receptor	**SLAM (CD150) on lymphocytes and nectin-4 on epithelium** — this explains the combination of profound immunosuppression with epithelial and respiratory signs in both	
Signs	Sudden high fever, serous then **mucopurulent oculo-nasal discharge matting the eyelids, necrotic erosive stomatitis** on gums, dental pad and tongue, foul-smelling **profuse diarrhoea**, bronchopneumonia, abortion	Biphasic fever, oculo-nasal discharge, **catarrhal then purulent conjunctivitis**, diarrhoea and vomiting, **hyperkeratosis of the nose and footpads, enamel hypoplasia** in survivors, and late **neurological disease — myoclonus (chorea), seizures, "old dog encephalitis"**
Lesions	**"Zebra striping" — linear haemorrhages on the mucosal folds of the large intestine and colon**; erosive lesions along the whole alimentary tract; congested lungs	Bronchopneumonia, **intracytoplasmic AND intranuclear eosinophilic inclusions** in bladder, bile duct and CNS; demyelination
Diagnosis	**c-ELISA and s-ELISA**, AGID, **RT-PCR on nasal/ocular swabs and lymph node**, isolation in Vero cells expressing SLAM (syncytia)	Cytoplasmic inclusions in conjunctival scrapings and buffy coat, **FAT**, RT-PCR, IgM ELISA, CSF antibody
Control	**Live attenuated Sungri/96 vaccine — a single dose gives lifelong immunity**; India's national PPR eradication programme; notifiable	Live attenuated (Onderstepoort, Rockborn) from 6–8 weeks, boosters till 16 weeks; recombinant canarypox-vectored vaccine

> **Morbillivirus family resemblance** Rinderpest (**eradicated globally in 2011** — the second disease after smallpox), PPR, canine distemper, measles and phocine distemper are all morbilliviruses; all cause fever + oculo-nasal discharge + erosive stomatitis + diarrhoea + immunosuppression + syncytia with inclusions. If you can write one, you can write them all.

Rhabdoviridae — rabies

SYSTEMATIC

- **Structure: bullet-shaped**, enveloped, helical, non-segmented (−)ssRNA, ~180 × 75 nm; five proteins **N, P, M, G, L**. The **G (glycoprotein) spike is the only one that induces neutralising antibody** and is therefore the vaccine and the virulence determinant; N protein is the target of the diagnostic FAT conjugate. **Genus *Lyssavirus***; a single serotype but seven-plus genotypes.
- **Pathogenesis — write the chain:** bite deposits saliva → replication in local **myocytes** → binds the **nicotinic acetylcholine receptor** at the neuromuscular junction → **centripetal, retrograde axonal transport at about 3 mm per hour** → spinal cord → brain, where it replicates massively → **centrifugal spread along nerves to the salivary glands, cornea, skin and other organs. There is no viraemia** — an important negative point. The incubation period (2 weeks to over a year) depends on the **distance of the bite from the brain** and the dose.

- **Clinical forms: Prodromal** (behaviour change, 2–3 days) → **Furious** (aggression, pica, roaming, hydrophobia in man, no fear) → **Paralytic/dumb** (drooping jaw, drooling, ascending paralysis, death). Cattle show bellowing, tenesmus and paralysis; horses colic and self-mutilation. **Invariably fatal once signs appear.**
- **Diagnosis (post-mortem only):** collect **hippocampus (Ammon's horn), cerebellum and brain stem. Direct FAT on brain impression smears is the WOAH gold standard. Seller's stain / haematoxylin-eosin for Negri bodies — intracytoplasmic, eosinophilic, in Purkinje and hippocampal neurons; present in only 50–80% of cases, so a negative Negri does not rule out rabies.** Also mouse inoculation test, **RTCIT** in BHK-21/N2a cells, RT-PCR, **dRIT** for field use. Antibody by **RFFIT or FAVN** for export certification (adequate titre ≥0.5 IU/ml).
- **Prophylaxis:** pre-exposure vaccination of dogs and of at-risk people. **Post-exposure prophylaxis** = immediate **washing of the wound with soap and running water for 15 minutes** + antiseptic, **rabies immunoglobulin infiltrated around the wound** for category III, and a **cell-culture vaccine** (PCECV, PVRV, HDCV) on the **Essen 0-3-7-14-28 intramuscular** or intradermal **Thai Red Cross** schedule. **Semple (sheep brain) vaccine is obsolete in India** because of neuroparalytic accidents. **Oral bait vaccination of wildlife** with vaccinia- or adenovirus-vectored G protein. Goal: **dog-mediated human rabies elimination by 2030**.
- **Bovine ephemeral fever** — also *Rhabdoviridae*, *Culicoides*- and mosquito-borne, "three-day sickness" of cattle: sudden high fever, stiffness, lameness, sudden drop in milk, recumbency, and rapid spontaneous recovery in about three days.

Orthomyxoviridae & Bornaviridae

SYSTEMATIC

- **Influenza A — structure:** enveloped, helical, **8 segments** of (−)ssRNA. Two surface glycoproteins: **HA (haemagglutinin — attachment to sialic acid, target of neutralising antibody, 18 subtypes)** and **NA (neuraminidase — releases progeny by cleaving sialic acid, 11 subtypes)**; plus the **M2 ion channel** (amantadine target) and internal **NP and M** proteins which define the type (A, B, C).
- **The two mechanisms of antigenic change — the single most asked point: Antigenic DRIFT** = gradual accumulation of point mutations in HA and NA, occurs in all types, causes seasonal epidemics and the need for annual vaccine updating. **Antigenic SHIFT** = sudden **reassortment** of whole segments between two strains co-infecting one cell, occurs only in type A, produces a completely new subtype against which the population has no immunity → **pandemic**. **The pig is the classic "mixing vessel"** because it bears both avian (α-2,3) and human (α-2,6) sialic-acid receptors.
- **Avian influenza:** classified by pathogenicity — **LPAI** versus **HPAI**, which is defined by an **IVPI > 1.2** or a **polybasic HA cleavage site**; HPAI is restricted to **H5 and H7** subtypes. HPAI **H5N1** causes sudden very high mortality, cyanotic combs and wattles, subcutaneous oedema of the head, haemorrhages on the shanks and visceral serosae. **Notifiable, zoonotic; the policy in India is stamping out, not vaccination.** Wild waterfowl are the natural reservoir.
- **Equine influenza** — H7N7 (rare now) and **H3N8**; harsh dry hacking cough, fever, nasal discharge; spreads explosively through racing stables; inactivated vaccines must be antigenically updated. **Swine influenza** — H1N1, H1N2, H3N2; high morbidity, low mortality, "thumps"; zoonotic potential.
- **Diagnosis:** egg or MDCK-cell isolation with **trypsin** in the medium; **HA and HI**; antigen-capture ELISA; **real-time RT-PCR on the M gene** for typing and on H5/H7 for subtyping; AGID for the type-specific NP antigen.
- **Borna disease virus** — *Bornaviridae*, the only (−)ssRNA virus that **replicates in the NUCLEUS**; non-cytolytic and persistent. Causes **"sad horse disease"** — a progressive immune-mediated meningoencephalomyelitis of horses and sheep in central Europe with behavioural change, ataxia and somnolence; **Joest-Degen intranuclear inclusion bodies** in hippocampal neurons. Also proventricular dilatation disease of parrots (avian bornavirus).

Sheet 03 · 60-second self-test

RECALL

Prompt	Answer
Target cell of IBD virus	Immature IgM^+ B cells of the bursa
Lasting damage of Gumboro	Immunosuppression
Vector of bluetongue and AHS	*Culicoides* midge
Pathognomonic lesion of bluetongue	Haemorrhage at base of pulmonary artery
Dunkop and dikkop forms	African horse sickness
Wheel-like virus of neonatal scour	Rotavirus
Two spikes of NDV	HN and F
Determinant of NDV virulence	F_0 cleavage site
Mesogenic Indian ND vaccine strain	R2B (Mukteswar)
ICPI of a velogenic strain	Approaches 2.0
Zebra striping of the colon	PPR
PPR vaccine strain used in India	Sungri/96
Receptor of morbilliviruses	SLAM (CD150) / nectin-4
Hard pad disease	Canine distemper
Shape of rabies virion	Bullet-shaped
Only rabies protein giving neutralising antibody	G glycoprotein
Speed of rabies axonal travel	About 3 mm per hour
Best brain sample for rabies	Hippocampus (Ammon's horn)
Antigenic shift occurs by	Reassortment of segments
Definition of HPAI	IVPI > 1.2 or polybasic HA cleavage site
Only (−)RNA virus replicating in nucleus	Borna disease virus
Joest-Degen bodies	Borna disease

Positive-Sense RNA Viruses & the Retroviruses

Picornaviridae — foot-and-mouth disease

SYSTEMATIC

- **Structure: naked** icosahedral, (+)ssRNA, ~30 nm — one of the smallest and, being naked, one of the most **environmentally stable** animal viruses. Capsid of **VP1, VP2, VP3 (surface) and VP4 (internal); VP1 carries the major immunogenic site and the RGD motif that binds the integrin receptor**. Genome also encodes the **non-structural proteins (3ABC etc.)** that are the basis of DIVA testing.
- **Seven serotypes — O, A, C, Asia-1, SAT-1, SAT-2, SAT-3 — with NO cross-protection**, and many subtypes within each. **In India: O, A and Asia-1 circulate; serotype C has not been recorded since 1995.** This is why the Indian vaccine is **trivalent (O, A, Asia-1)**.
- **Extremely labile to pH** — inactivated below pH 6.5 and above pH 11. Hence **sodium carbonate (4%), citric acid (2%) and sodium hydroxide** are the disinfectants of choice, while ordinary phenolics fail. This also explains why the virus survives in **chilled but not in properly acidified (rigor-matured) meat**.
- **Hosts:** all **cloven-hoofed** animals — cattle, buffalo, sheep, goat, pig, wildlife. **Horses are refractory** (a classic one-mark question). Pigs are amplifiers that excrete enormous amounts of aerosol; sheep are silent maintainers.
- **Signs and lesions:** fever, then **vesicles/aphthae on the tongue, dental pad, gums, muzzle, interdigital space, coronary band and teats**, which rupture to leave raw erosions; profuse **ropy salivation, smacking of the lips**, lameness, sharp drop in milk. Morbidity approaches 100%, adult mortality is low, but **mortality in young stock is high because of "tiger heart" — myocarditis with grey-yellow streaks in the myocardium**. Sequelae: chronic **"panting" syndrome** with dyspnoea and hair-coat changes, and lifelong **carrier state in the oropharynx** of ruminants.
- **Diagnosis:** the best sample is **vesicular epithelium (tongue/foot) in 50% glycerol-phosphate buffer, pH 7.2–7.6, cold chain**; also oesophageal-pharyngeal fluid by **probang cup** for carriers. Tests: **sandwich ELISA for typing (the routine)**, CFT, isolation in BHK-21 or primary bovine thyroid cells, inoculation of unweaned mice, **real-time RT-PCR**, and **NSP-ELISA (3ABC) to distinguish infected from vaccinated animals**.
- **Control:** inactivated trivalent oil/alum vaccine, first dose at 4 months, booster at 1 month, then **six-monthly**; India's **FMD-CP / NADCP** mass vaccination with ear-tagging and INAPH recording. Ring vaccination, movement control, disinfection, and stamping out where policy permits. **Notifiable and transboundary.**

> Differentiate the vesicular diseases — FMD, vesicular stomatitis (rhabdovirus; affects horses, which rules it apart from FMD), swine vesicular disease (enterovirus, pigs only) and vesicular exanthema of swine (calicivirus). Clinically indistinguishable in pigs — only the laboratory can differentiate them, and that sentence is worth a mark.

Corona-, Arteri-, Calici- & Togaviridae

SYSTEMATIC

- ***Coronaviridae*** — enveloped, helical, (+)ssRNA, the **largest RNA genome (~30 kb)**, with club-shaped **S (spike) peplomers giving the crown or solar-corona appearance**. High rate of **recombination** and mutation.
- **Infectious bronchitis virus (IBV)** — the most contagious poultry virus; **respiratory** (gasping, tracheal rales, snicking), **renal** (nephropathogenic strains → urate deposits, "gout") and **reproductive** (permanent oviduct damage in chicks infected under 2 weeks → **false layers**; in layers, a fall in production with **soft-shelled, misshapen eggs and watery albumen**) forms. Very many serotypes with poor cross-protection — **vaccine must match the local serotype (Massachusetts H120, H52, 4/91)**.
- **Transmissible gastroenteritis virus (TGEV)** of pigs — destroys the **villous enterocytes of the jejunum and ileum** → severe villous atrophy → profuse watery diarrhoea and vomiting with **near 100% mortality in piglets under 2 weeks** and mild disease in adults. Control by **feed-back/oral exposure of pregnant sows to generate lactogenic IgA**. **PRCV** is a spike-deletion mutant that is respiratory and confers cross-protection. Related: PEDV, bovine coronavirus (calf scour and winter dysentery), **FIP virus** (a mutant of feline enteric coronavirus; immune-complex and ADE-driven, with effusive "wet" and granulomatous "dry" forms).
- ***Arteriviridae* — equine viral arteritis.** Causes fever, **limb and ventral oedema, supraorbital/periorbital oedema, conjunctivitis ("pink eye")**, urticaria, and **abortion at any stage**. The critical epidemiological point: **the virus persists in the accessory sex glands of the stallion, which becomes a long-term shedder in semen** — so screening and certification of breeding stallions is the control measure. Modified live and inactivated vaccines exist; test before vaccinating so serology remains interpretable. **PRRS virus** is in the same family — "blue ear disease", reproductive failure in sows and respiratory disease in piglets, targeting **alveolar macrophages**.
- ***Caliciviridae* — feline calicivirus.** Naked, (+)ssRNA, **cup-shaped depressions** on the surface. Part of the feline upper-respiratory complex with FHV-1; the hallmark is **oral and lingual ULCERATION** (FHV-1 gives more ocular signs), plus a **limping/lameness syndrome** and virulent systemic strains. Naked, so it is **resistant to alcohol — hypochlorite is needed**; cats become carriers shedding oropharyngeally. Also in the family: **rabbit haemorrhagic disease virus** and vesicular exanthema of swine.
- ***Togaviridae* — equine encephalomyelitis (EEE, WEE, VEE).** Mosquito-borne **arboviruses** maintained in a **bird–mosquito cycle**; the **horse and man are dead-end hosts** except in VEE, where equines do amplify. Biphasic fever, then encephalitis with depression ("sleeping sickness"), compulsive walking, head pressing, paralysis. **EEE is the most severe (mortality up to 90%), WEE mild-to-moderate, VEE intermediate but epidemic.** All are **zoonotic**; diagnosis by IgM-capture ELISA, HI, RT-PCR and virus isolation from brain. Control by inactivated vaccine and mosquito control.

Flaviviridae — classical swine fever & BVD

TABLE

	Classical swine fever (hog cholera)	Bovine viral diarrhoea / mucosal disease
Genus	*Pestivirus* — enveloped, (+)ssRNA; the genus also holds border disease of sheep. Note that **pestiviruses do NOT have an arthropod vector** despite being flaviviruses	
Key biology	**Endotheliotropic and lymphotropic** → widespread haemorrhage and lymphoid depletion; a single serotype	Two **biotypes — non-cytopathic (ncp) and cytopathic (cp)**, and two genotypes (BVDV-1, BVDV-2). **The ncp biotype is the one that establishes persistent infection**
Signs	High fever with huddling, conjunctivitis, constipation then diarrhoea, incoordination and a staggering "wobbly" hindquarter gait, **purple discoloration of the ears, abdomen and inner thigh**; transplacental infection gives **congenital tremor, mummification and stillbirth**	Usually subclinical; acute BVD gives fever, oral erosions, diarrhoea, leucopenia and **immunosuppression**; reproductive losses — early embryonic death, abortion, **cerebellar hypoplasia and cataract** if infected at 100–150 days
Signature lesion	**"Turkey-egg" kidney — petechial haemorrhages on a pale cortex**; **button ulcers in the colon and at the ileocaecal valve**; infarcted enlarged mottled spleen; haemorrhagic lymph nodes; encephalitis with perivascular cuffing	**Mucosal disease** — erosions and ulcers of the whole alimentary tract, **linear oesophageal erosions**, necrosis of Peyer's patches, interdigital erosions
The concept to explain	Chronic and late-onset forms mean apparently healthy carriers spread the virus; feeding of uncooked swill is the classic source	**PERSISTENTLY INFECTED (PI) ANIMAL** — a foetus infected with the **ncp** biotype **before day 125** of gestation, i.e. before immunocompetence, recognises the virus as self, is **immunotolerant, seronegative but virus-positive for life**, and sheds continuously. If that PI animal is later superinfected with a **cp** biotype arising by mutation, it develops fatal **mucosal disease**. **Eradication depends on finding and culling PI animals** — the highest-yield single concept in this topic
Diagnosis	FAT on tonsil and kidney, antigen ELISA, RT-PCR, isolation in PK-15 cells (**non-cytopathic — needs**	**Antigen-capture ELISA or RT-PCR on ear-notch skin**, repeated after 3 weeks to confirm persistence; virus isolation in MDBK cells; antibody ELISA (a PI

	Classical swine fever (hog cholera)	Bovine viral diarrhoea / mucosal disease
	FAT to read), **END and exaltation of Newcastle disease virus test**	animal is **antibody-negative**)
Control	Live attenuated **lapinised Chinese C strain** (used in India), cell-culture vaccine; stamping out and swill-feeding ban in free countries; **notifiable**	Test-and-cull PI animals + biosecurity + killed or MLV vaccination of breeding stock before service

Retroviridae

SYSTEMATIC

- **Defining feature:** enveloped, **two identical copies of (+)ssRNA** (diploid) plus **reverse transcriptase**. Replication: RNA → **cDNA by RT** → dsDNA → **integrated into the host chromosome by integrase as a PROVIRUS** → transcribed by host RNA polymerase II. This integration is why retroviral infection is **lifelong** and why these viruses cause cancer.
- **Three essential genes — *gag* (core proteins), *pol* (RT, protease, integrase), *env* (envelope glycoproteins)**, flanked by **LTRs** that act as promoters. Lentiviruses carry additional regulatory genes.
- **Bovine leukaemia virus (enzootic bovine leucosis)** — *Deltaretrovirus*. Long incubation; most cattle are seropositive and healthy, about 30% develop persistent lymphocytosis, and only 1–5% develop **B-cell lymphosarcoma** in the abomasum, heart (right atrium), uterus, spinal canal and retrobulbar space. Transmitted **iatrogenically by blood — needles, dehorning, rectal sleeves**, and by biting flies and colostrum. Diagnosis by **AGID and ELISA on serum or bulk milk, plus PCR for provirus**. No vaccine; control is test-and-cull with hygiene. Distinguish from **sporadic bovine leucosis** (juvenile, thymic and cutaneous forms) which is **not** BLV-associated.
- **Avian leukosis/sarcoma group** — subgroups A–J; **lymphoid leucosis affects birds over 16 weeks, with bursal-dependent B-cell tumours and a nodular liver ("big liver disease")**. **Vertically transmitted through the egg**, so control is by eradication from breeder flocks, not vaccination. Rous sarcoma virus (*v-src*) is the historic transforming member.
- **Lentiviruses — "slow viruses": Visna-maedi** of sheep (*maedi* = progressive interstitial pneumonia and dyspnoea; *visna* = wasting demyelinating encephalitis; also indurative mastitis, "hard bag"); **caprine arthritis-encephalitis virus** (carpal arthritis in adults, leucoencephalomyelitis in kids, spread mainly through **milk and colostrum**); **equine infectious anaemia** — "swamp fever", transmitted by **tabanid horse flies and contaminated needles**, cyclical fever, anaemia and thrombocytopenia from **antigenic variation with immune-complex and immune-mediated haemolysis**, lifelong carrier, diagnosed by the **Coggins (AGID) test**, and controlled by test-and-slaughter or lifelong isolation — **there is no vaccine and no treatment**. Also FIV and BIV.

Distinguish the three lymphoid tumours of poultry — Marek's disease (herpesvirus, birds 6–20 weeks, T cells, enlarged peripheral nerves, iris depigmentation, vaccine available) · Lymphoid leucosis (retrovirus, birds over 16 weeks, B cells, bursa-dependent, no nerve enlargement, no vaccine) · Reticuloendotheliosis. That three-way table appears in the paper regularly.

Sheet 04 · 60-second self-test

RECALL

FMD serotypes	O, A, C, Asia-1, SAT-1, 2, 3 — no cross-protection
Serotypes circulating in India	O, A, Asia-1
FMD immunogenic capsid protein	VP1
FMD DIVA marker	Non-structural protein 3ABC
Transport medium for FMD epithelium	50% glycerol-phosphate buffer pH 7.4
Tiger heart in young stock	FMD myocarditis
Species refractory to FMD	Horse
Vesicular disease affecting horses	Vesicular stomatitis
Largest RNA genome	Coronavirus (~30 kb)
False layers in poultry	Infectious bronchitis virus
Villous atrophy with 100% piglet mortality	TGE virus
Virus persisting in the stallion's sex glands	Equine viral arteritis
Blue ear disease	PRRS virus
Oral ulceration in cats	Feline calicivirus
Turkey-egg kidney and button ulcers	Classical swine fever
CSF vaccine strain used in India	Lapinised Chinese C strain
Biotype causing persistent infection	Non-cytopathic BVDV
PI animal is	Virus-positive, antibody-negative for life
Gestational window for BVDV PI	Before day 125
Three genes of a retrovirus	*gag*, *pol*, *env*
Integrated retroviral DNA	Provirus
Coggins test diagnoses	Equine infectious anaemia
Maedi and visna	Pneumonia and encephalitis of sheep
Lymphoid leucosis affects birds over	16 weeks — B cells, bursa-dependent

05 Large DNA Viruses — Pox, Asfar & Herpes

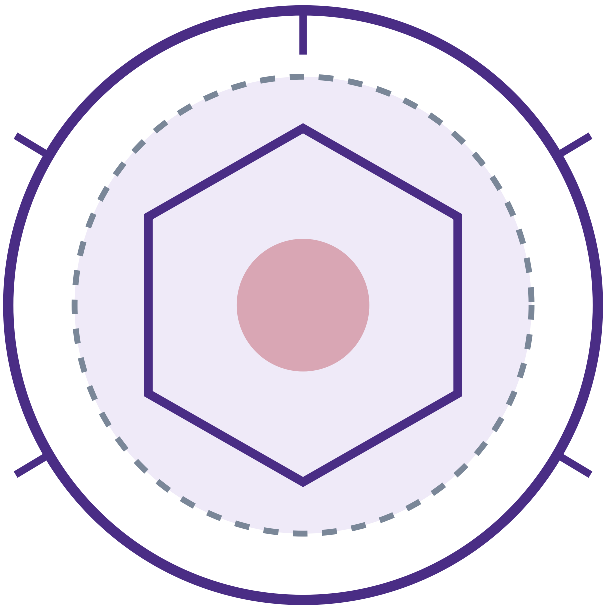
Poxviridae

SYSTEMATIC

- **Structure:** the **largest and most complex animal virus** (200–450 nm), **brick- or ovoid-shaped** with a **dumbbell-shaped core and two lateral bodies**; dsDNA. **Unique among DNA viruses in replicating entirely in the CYTOPLASM** (in "viral factories"), which is why it must carry its own transcriptase. **Very resistant to drying** — survives for months in scabs and wool.
- **Lesion sequence to write: macule → papule → vesicle → pustule → crust/scab → scar**. Pox lesions are epitheliotropic with ballooning degeneration and **intracytoplasmic eosinophilic inclusion bodies**.
- **Capripoxvirus — sheep pox, goat pox and lumpy skin disease. The most serious pox of livestock, notifiable and transboundary.** Sheep and goat pox: fever, generalised skin nodules especially on the woolless areas — muzzle, eyelids, axilla, groin, udder — plus **internal pox lesions in the lung as sharply circumscribed grey-white nodules. Lumpy skin disease** of cattle: high fever, **multiple firm intradermal nodules 2–5 cm over the whole body** that become necrotic "sit-fast" plugs, marked **lymph node enlargement**, limb oedema, drop in milk, and **vector-borne spread by biting flies, mosquitoes and ticks** — this vector transmission is what made LSD spread across Asia. Diagnosis: EM, histopathology, **PCR**, virus isolation on lamb testis cells or CAM. Control: live attenuated homologous vaccines; the **Kenyan sheep and goat pox strain and Neethling strain are used against LSD**.
- **Avipoxvirus — fowl pox.** Two forms: **cutaneous ("dry")** — wart-like nodules on the comb, wattle and unfeathered skin, spread by mosquitoes and by contact through skin abrasions, low mortality; and **diphtheritic ("wet")** — yellow caseous plaques in the mouth, pharynx and trachea causing asphyxia and higher mortality. **Bollinger bodies** (intracytoplasmic inclusions) with **Borrel bodies** (elementary particles) inside them. Vaccinated by **wing-web stab** with a live pigeon-pox or fowl-pox vaccine; check for a "take" after 7 days.
- **Orthopoxvirus — cowpox, buffalopox and vaccinia.** Cowpox causes **teat and udder lesions in cows and generalised disease in cats (rodent reservoir); zoonotic**. **Buffalopox** is important in India — lesions on the teats and udder of buffaloes with **painful ulcers on the hands of milkers**. Vaccinia is the smallpox vaccine strain and now the workhorse **viral vector**.
- **Parapoxvirus** — **orf (contagious ecthyma / contagious pustular dermatitis)** of sheep and goats: proliferative scabby lesions at the commissures of the lips and on the muzzle of lambs, and on the teats of ewes; **ovoid virion with a criss-cross "ball of wool" surface pattern on EM; zoonotic**. Also bovine papular stomatitis and pseudocowpox (milker's nodule).
- **Suipoxvirus** — swine pox, transmitted by the **pig louse *Haematopinus suis***. **Myxomatosis** (*Leporipoxvirus*) of rabbits, mosquito-borne, is the classic virus-host co-evolution example.

Herpesviridae — the family rules first

MECHANISM



Envelope + glycoprotein spikes

Tegument — protein layer, uni

Icosahedral capsid, 162 capso

Core — linear dsDNA around a

THE FAMILY RULE

Every herpesvirus establishes l

and reactivates under stress or

Subfamily	Behaviour & latency site	Members
Alphaherpesvirinae	Fast growth, wide host range, cytolytic; **latent in SENSORY GANGLIA**	BHV-1 (IBR/IPV), **equine herpesvirus 1 & 4**, pseudorabies (SHV-1), **ILTV**, duck plague, FHV-1, canine herpesvirus
Betaherpesvirinae	Slow growth, narrow host range, **cytomegaly**; latent in secretory glands and kidney	Cytomegaloviruses, porcine inclusion body rhinitis virus
Gammaherpesvirinae	**Lymphotropic and often oncogenic**; latent in lymphocytes	**Marek's disease virus** (biologically alpha but genomically gamma), malignant catarrhal fever viruses (OvHV-2, AlHV-1), Epstein-Barr virus

Cowdry type A The eosinophilic **intranuclear** inclusion with a clear halo, seen in herpesvirus infections — the histopathological signature to name in any herpes answer.

The veterinary herpesviruses

TABLE

Virus & disease	Clinical picture	The point that scores
BHV-1 — IBR / IPV	**Respiratory:** "red nose", fever, mucopurulent nasal discharge, tracheitis, conjunctivitis. **Genital:** infectious pustular vulvovaginitis and balanoposthitis. Also **abortion in the last trimester** with a foetus showing	**Latent in the trigeminal (respiratory) and sacral (genital) ganglia; reactivated experimentally by dexamethasone.** Any seropositive animal is a lifelong carrier — hence **test-and-cull**, and the availability of **gE-deleted marker (DIVA) vaccines** that allow eradication while vaccinating. Semen must be screened before AI

Virus & disease	Clinical picture	The point that scores
	pinpoint necrotic foci in the liver, and encephalitis in calves	
EHV-1 & EHV-4	EHV-4 mainly **rhinopneumonitis** in young horses; **EHV-1** causes rhinopneumonitis plus **abortion "storms" in the last third of gestation**, neonatal foal death, and **equine herpesvirus myeloencephalopathy** (ataxia, urinary incontinence, paresis)	Aborted foetus is **expelled fresh with the membranes**, showing **pulmonary oedema, hepatic necrotic foci and intranuclear inclusions** — the classic gross picture. Vaccinate mares at 5, 7 and 9 months of gestation
Marek's disease virus `GaHV-2, serotype 1`	**T-cell lymphoma** of chickens 6–20 weeks. **Classical form:** unilateral leg paralysis with one leg forward and one back, **enlarged, greyish, oedematous peripheral nerves (sciatic, brachial, vagus) that lose their cross-striations. Ocular form:** "grey eye" — iris depigmentation and irregular pupil. **Acute form:** visceral lymphomas in liver, gonad, spleen, kidney. **Cutaneous form:** feather-follicle nodules	**The virus is shed as fully infectious enveloped virus ONLY from the feather-follicle epithelium**, so dander is the source and the disease spreads by inhalation of dust. **The first neoplastic disease controlled by a vaccine** — HVT (turkey herpesvirus, serotype 3), SB-1 (serotype 2) and **CVI988/Rispens**, given **at day-old or in ovo at 18 days**. The vaccine prevents the tumour but not infection or shedding
ILTV `infectious laryngotracheitis`	Severe dyspnoea with **neck extension and gasping, expectoration of blood-stained mucus**, sudden drop in egg production. Haemorrhagic tracheitis with a caseous plug	**Syncytia with Cowdry type A intranuclear inclusions** in tracheal epithelium — diagnostic. Vaccinated by eye drop or drinking water only in endemic areas, since vaccine strains can revert and spread
Duck plague `duck viral enteritis`	Sudden high mortality in ducks, **prolapsed penis in drakes**, photophobia, greenish watery diarrhoea	**Annular bands of haemorrhage in the intestine, "duck plague" oesophageal and cloacal button-like plaques**, and necrotic foci in the liver. Wild waterfowl are carriers; live attenuated vaccine
Pseudorabies `Aujeszky's disease, SHV-1`	**Pig is the natural host and the reservoir** — piglets show fatal CNS disease, sows abort, adults show mild respiratory signs. In **dead-end hosts (cattle, dog, cat, sheep)** it causes **"MAD ITCH" — intense localised pruritus with self-mutilation, followed by death within 48 h**	The mad-itch picture in a cow on a pig farm is the exam vignette. **gE-deleted DIVA vaccine** is the basis of national eradication. **Not** caused by rabies virus despite the name — say so
Malignant catarrhal fever	Sporadic, invariably fatal in cattle and buffalo. High fever, bilateral **corneal opacity beginning at the limbus and moving centrally**, severe mucopurulent nasal and ocular discharge, erosions of the muzzle and oral cavity, generalised lymphadenopathy, nervous signs	**Two epidemiological forms: wildebeest-associated (AlHV-1)** in Africa, and **sheep-associated (OvHV-2)** worldwide, in which **sheep are inapparent carriers and cattle are dead-end hosts. Cattle-to-cattle transmission does not occur**, so control is simply **separating cattle from lambing sheep**. Histology: **lymphoproliferative vasculitis with fibrinoid necrosis**. Diagnosis by PCR; no vaccine, no treatment

Asfarviridae — African swine fever

SYSTEMATIC

- **Unique standing:** ASFV is the **only DNA arbovirus** and the **only member of its family**; a large icosahedral dsDNA virus that, like poxvirus, replicates chiefly in the **cytoplasm**. It targets **monocytes and macrophages**.
- **Epidemiology — the reason it is so feared:** the sylvatic cycle runs between **warthogs and bushpigs (inapparent reservoirs)** and ***Ornithodoros* soft ticks**, which are both vector and reservoir. Domestic pigs are infected by tick bite, by contact, or by **feeding of uncooked swill containing pork products** — the virus survives for months in chilled, frozen and cured meat. **There is no vaccine and no treatment.**
- **Clinical and lesions:** peracute to chronic. High fever, **reddening/cyanosis of the ears, snout, abdomen and limbs**, huddling, ataxia, abortion, and mortality approaching 100% in naïve herds. Post mortem: **massively enlarged, friable, dark "infarcted" spleen, markedly haemorrhagic gastrohepatic and renal lymph nodes that look like blood clots**, petechiae throughout the kidney cortex, hydropericardium, pulmonary oedema.
- **Diagnosis:** clinically **indistinguishable from classical swine fever** — this is the single most important differential and the laboratory must decide. Use **PCR (the routine), antigen ELISA, FAT on tissue smears, and haemadsorption in porcine leucocyte cultures, in which erythrocytes rosette around infected macrophages ("rosette formation")**. Antibody ELISA and IB for chronic cases (unlike CSF, ASF survivors do make antibody, but it is **not protective**).
- **Control:** strict biosecurity, **ban on swill feeding**, movement control, stamping out with safe carcass disposal, tick control, and separation from wild suids. **Notifiable and transboundary; ASF entered India in 2020 and is now the major threat to the piggery sector in the North-East.**

> ASF vs CSF in one line each: ASF is a DNA virus, tick-borne, with a huge friable haemorrhagic spleen and haemorrhagic nodes, and no vaccine. CSF is an RNA flavivirus, no vector, with an infarcted (not enlarged) spleen, turkey-egg kidney and button ulcers, and an effective live vaccine.

06 Small DNA Viruses, Prions & Emerging Infections

t28–t33

Adeno-, Papilloma-, Parvo- & Circoviridae

SYSTEMATIC

- ***Adenoviridae*** — **naked icosahedral dsDNA with a fibre projecting from each penton**, the attachment organ. **Infectious canine hepatitis (canine adenovirus 1)**: fever, abdominal pain, tonsillitis, hepatic necrosis with a **markedly oedematous gall-bladder wall**, petechiae, and the immune-complex sequel **"blue eye" — corneal oedema appearing 1–3 weeks after infection or after live CAV-1 vaccination**, which is why modern vaccines use **CAV-2**, giving cross-protection without blue eye. **Basophilic intranuclear inclusions in hepatocytes and endothelium**; virus is shed in urine for up to **six months**. **CAV-2** itself causes infectious tracheobronchitis (kennel cough). **Egg drop syndrome 1976 (EDS-76)** — a duck adenovirus in hens producing **thin-shelled, soft-shelled and shell-less eggs with no other signs**; the virus **haemagglutinates fowl RBCs**, so it is diagnosed by **HI**. **Fowl adenovirus** — inclusion body hepatitis and **hydropericardium syndrome (Angara disease/litchi heart)** in broilers 3–5 weeks, with straw-coloured fluid in the pericardium and a swollen mottled liver.
- ***Papillomaviridae*** — naked icosahedral **circular dsDNA**, strictly **species- and site-specific**, **epitheliotropic**, and **cannot be grown in cell culture**. Causes benign **warts (papillomas)** that usually regress spontaneously through cell-mediated immunity. **Bovine papillomatosis** — cauliflower-like warts on head, neck, teats and penis of young cattle; **BPV-1 and BPV-2 also infect the horse to produce the EQUINE SARCOID**, the commonest equine skin tumour, which is **locally aggressive and does not regress**. **BPV-4 with bracken-fern immunosuppression causes alimentary and urinary bladder carcinoma (enzootic haematuria)**. Oncogenesis is by **E6 and E7 proteins inactivating p53 and Rb**. Treat by autogenous wart vaccine, surgical removal or crushing (which stimulates immunity).
- ***Parvoviridae*** — the smallest DNA viruses, **naked, ssDNA**, therefore **extremely resistant** (survive a year in the environment; require **hypochlorite**, not alcohol) and **dependent on host DNA polymerase, so they replicate only in ACTIVELY DIVIDING cells** — this single fact explains their entire pathology. **Canine parvovirus 2** (variants 2a, 2b, 2c) attacks the **crypt cells of the intestine and the bone marrow → haemorrhagic enteritis with foul, blood-stained diarrhoea, vomiting, rapid dehydration and leucopenia** in pups 6 weeks to 6 months, and **myocarditis in pups infected under 8 weeks or in utero**. Diagnosis: faecal antigen ELISA/lateral flow, HA with pig RBCs, PCR. Vaccinate from 6 weeks, repeating till 16 weeks to overcome maternal antibody. **Feline panleukopenia** — the same biology in cats: enteritis with profound **leucopenia**, and infection *in utero* or in the neonate destroys the dividing external granular layer of the cerebellum → **cerebellar hypoplasia with ataxia and intention tremor**. **Porcine parvovirus** → the **SMEDI** syndrome: stillbirth, mummification, embryonic death and infertility, with mummified foetuses **of different sizes in one litter** and no illness in the sow.
- ***Circoviridae*** — the **smallest animal viruses (17–22 nm), naked, circular ssDNA**. **Chicken anaemia virus** (a gyrovirus) infects **haemocytoblasts in the marrow and T-cell precursors in the thymus → aplastic anaemia with watery blood and a PCV below 20, thymic and bursal atrophy, and gangrenous dermatitis** in chicks 2–4 weeks; it is **vertically transmitted through the egg** and its main cost is immunosuppression and vaccine failure — breeders are vaccinated to give maternal protection. **Porcine circovirus 2** is the cause of **PCVAD/PMWS — post-weaning multisystemic wasting syndrome**: progressive wasting, pallor, jaundice, dyspnoea and enlarged lymph nodes in pigs 6–14 weeks, plus PDNS and reproductive failure; it too is profoundly **immunosuppressive**, with basophilic botryoid (grape-like) cytoplasmic inclusions in macrophages. Highly effective PCV2 vaccines exist.

Prions — transmissible spongiform encephalopathies

MECHANISM

PrP^C — normal

Host gene *PRNP*, chromosom
α-helical · protease-SENSITIV

→

PrP^{Sc} — disease form

β-pleated sheet · proteas
insoluble · aggregates int

PrPSc acts as a template — autocata

NO nucleic acid → no immune response, no fever, no inflar
no interferon, and **resistant to formalin, autoclaving at 12**
irradiation. Destroyed only by 1 N NaOH or 134 °C for 18 m

- **Common features of all TSEs:** very long incubation (years), invariably **fatal**, **no inflammatory or immune response**, and a characteristic histopathology of **spongiform vacuolation of the grey-matter neuropil, neuronal loss, astrocytosis and amyloid plaques, with no lesion outside the CNS**.
- **Scrapie** — sheep and goats. Signs: intense **pruritus with wool loss from rubbing**, a **"nibble reflex"**, ataxia, tremor and progressive wasting over months. Transmitted **horizontally from the placenta and birth fluids at lambing** and vertically. **Susceptibility is governed by PRNP codons 136, 154 and 171 — the ARR/ARR genotype is highly resistant, VRQ is susceptible**, so control is by **genotyping and breeding for resistance**.
- **Bovine spongiform encephalopathy ("mad cow disease")** — arose from feeding **meat-and-bone meal containing infected ruminant tissue**; incubation 4–6 years. Signs: apprehension, hyperaesthesia to sound and touch, kicking in the parlour, ataxia and falling — hence "mad cow". **Controlled by the ruminant feed ban and by removal of specified risk material** (brain, spinal cord, distal ileum, eyes, tonsil) from the food chain. **Zoonotic — the cause of variant Creutzfeldt-Jakob disease in man**, which is the reason for the global response.
- **Others to name: chronic wasting disease** of deer and elk (the only TSE spreading efficiently in a wild population), transmissible mink encephalopathy, feline spongiform encephalopathy; in man kuru, CJD, GSS and fatal familial insomnia.
- Diagnosis — **post mortem only: histopathology of the brain stem at the obex** for spongiform change; **immunohistochemistry for PrP^{Sc}**; **Western blot after proteinase K digestion — the confirmatory test**; rapid ELISA for surveillance screening; detection of **scrapie-associated fibrils** by EM; RT-QuIC as the modern amplification assay. **There is no ante-mortem test in routine use, no treatment and no vaccine.**

Emerging, re-emerging & transboundary viral infections

RAPID-FIRE

Why viruses emerge	High mutation rate of RNA viruses and **reassortment** in segmented ones; **expansion of the wildlife-livestock-human interface** through deforestation and land-use change; intensification of livestock production; the live-animal and bushmeat trade; **climate change extending vector ranges**; global travel and trade; and better detection technology, which makes some "emergence" apparent rather than real

Key statistic to quote	About **60–75% of emerging infectious diseases of humans are zoonotic**, and the majority of those originate in **wildlife**, with **bats and rodents** the most important reservoirs
Bat-origin viruses	**Nipah virus** (*Paramyxoviridae, Henipavirus*) — fruit bats (*Pteropus*) → **pigs as amplifiers in Malaysia; in India (Siliguri 2001, Kerala 2018 onward) direct bat-to-human spread through contaminated date-palm sap**, causing encephalitis with very high case fatality; BSL-4 agent. **Hendra virus** — bats → horses → man in Australia. Also SARS, MERS and rabies-related lyssaviruses, Ebola and Marburg
Arboviral expansion	**Bluetongue** moving north in Europe, **lumpy skin disease** spreading across Asia including India since 2019, **Rift Valley fever**, West Nile virus, Japanese encephalitis (pigs as amplifiers, *Culex* vector, a major zoonosis in India), Crimean-Congo haemorrhagic fever (*Hyalomma* ticks, livestock handlers)
Re-emerging in India	Lumpy skin disease, African swine fever (from 2020), highly pathogenic avian influenza H5N1, PPR in sheep and goats, canine parvovirus variants, glanders, rabies
Transboundary animal diseases (TADs)	FMD, PPR, CSF, ASF, HPAI, Newcastle disease, bluetongue, LSD, sheep and goat pox, African horse sickness, Rift Valley fever, CBPP — all **WOAH-listed and notifiable**, with major trade consequences
The One Health response	Integrated **human, animal and environmental** surveillance; the **WOAH-FAO-WHO-UNEP quadripartite**; risk analysis and early-warning systems (GLEWS); **disease-free zoning and compartmentalisation**; biosecurity, quarantine, movement control and ring vaccination; genomic surveillance and phylogenetic outbreak tracing; capacity for **BSL-3/BSL-4** containment; and in India the **NADCP**, NCDC-ICAR joint surveillance and the National One Health Mission

Two skeletons that cover the virology paper

SKELETON

A · "Write about *[any virus]*" — 8-heading frame

1. **Classification** — family, genus, genome type, symmetry, envelope
2. **Morphology & physicochemical properties** — size, shape, resistance, the one signature feature
3. **Antigenic structure** — serotypes and the protein that induces neutralising antibody
4. **Epidemiology** — hosts, reservoir, vector, route, season, distribution in India
5. **Pathogenesis** — entry, primary replication, viraemia, target organ
6. **Clinical signs and gross & microscopic lesions** — lead with the pathognomonic one
7. **Laboratory diagnosis** — sample & transport, direct detection, isolation system, serology, molecular test
8. **Immunity, vaccine and control** — name the strain used in India; close with the notifiable/zoonotic status

B · "Laboratory diagnosis of a viral disease" — 6-step frame

1. **Sample** — correct tissue, collected early in the febrile phase, in **viral transport medium with antibiotics, on ice**; brain in 50% glycerol saline for rabies, epithelium in glycerol-phosphate buffer for FMD
2. **Direct detection** — electron microscopy, **FAT/immunoperoxidase**, antigen-capture ELISA, inclusion bodies in histopathology
3. **Isolation** — embryonated egg by the appropriate route, cell culture reading the **CPE, haemadsorption or plaques**, or animal inoculation
4. **Identification** — neutralisation, HI, ELISA, immunofluorescence on the isolate
5. **Serology** — paired sera, **a four-fold rise or IgM detection** proves recent infection; SNT, HI, AGID, CFT, ELISA
6. **Molecular** — RT-PCR/real-time PCR, sequencing for genotyping, and **DIVA testing where a marker vaccine is in use**

Sheets 05–06 · final sweep

RECALL

Only DNA viruses in the cytoplasm	Pox and African swine fever
Bollinger and Borrel bodies	Fowl pox
Vaccine strain for lumpy skin disease	Kenyan sheep-goat pox / Neethling
Ball-of-wool virion on EM	Orf (parapoxvirus)
Pox transmitted by *Haematopinus suis*	Swine pox
Protein layer unique to herpesviruses	Tegument
IBR latency site	Trigeminal ganglion
Marker vaccine for IBR and Aujeszky's	gE-deleted
Site of infectious Marek's virus shedding	Feather follicle epithelium
First cancer prevented by vaccination	Marek's disease (HVT, CVI988)
Mad itch in cattle	Pseudorabies (Aujeszky's)
Corneal opacity from limbus inward	Malignant catarrhal fever
Reservoir of sheep-associated MCF	Sheep (OvHV-2); cattle are dead-end
Only DNA arbovirus	African swine fever virus
Rosette/haemadsorption test	ASF in porcine leucocyte culture
Blue eye after vaccination	CAV-1 — hence CAV-2 vaccines
Egg drop syndrome is diagnosed by	HI (virus haemagglutinates fowl RBC)
Angara disease / litchi heart	Hydropericardium, fowl adenovirus
Papillomavirus oncoproteins	E6 and E7 → inactivate p53 and Rb
Equine sarcoid is caused by	BPV-1 and BPV-2
Parvoviruses need which cells	Actively dividing cells
Cerebellar hypoplasia in kittens	Feline panleukopenia virus
SMEDI syndrome	Porcine parvovirus
Smallest animal virus, aplastic anaemia	Chicken anaemia virus (circovirus)

PMWS in weaner pigs	Porcine circovirus 2
Normal and disease prion protein	PrP^{C} α-helical · PrP^{Sc} β-sheet
Gene coding prion protein	*PRNP*
Confirmatory test for BSE/scrapie	Western blot after proteinase K
Best brain site for TSE histology	Obex of the brain stem
Origin of BSE	Ruminant meat-and-bone meal in feed
Nipah virus reservoir	*Pteropus* fruit bats
Amplifier host of Japanese encephalitis	Pig

`33 / 33 syllabus topics` `Paper II · Unit 5` `Full pass ≈ 30 min` `Night-before: sheets 03–06 self-tests + the two skeletons`